SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Concept Mapping
Weightage:	35%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	CSA1214/2025	Date:	17/07/26

Instructions to Students

- Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.
- Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.
- Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases.
- Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.
- Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

QUESTIONS

1. A software company is upgrading its office computers and must decide whether to use ARM Cortex, Intel Core, or AMD Ryzen processors for different workloads. Construct a concept map linking these modern architectures with their instruction execution, cache usage, processing capability, and application areas. Show how their architectural features influence performance and energy efficiency. How do modern processor architectures differ in meeting the requirements of mobile, desktop, and high-performance computing systems?

2. A computer technician is troubleshooting a system that stores machine instructions in memory but displays memory addresses in a debugger. Construct a concept map linking Binary and Hexadecimal Number Systems with instruction representation, memory storage, CPU interpretation, and data conversion. Show how instructions are represented, stored, and interpreted during execution. How do binary and hexadecimal number systems simplify memory organization, instruction representation, and debugging?

3. A manufacturing company is upgrading its automated inspection system and notices delays when multiple sensors and storage devices communicate with the processor simultaneously. Construct a concept map linking CPU, Memory, I/O Devices, Single-bus, Multi-bus, and Hierarchical Bus structures, showing how data and control signals are transferred. Illustrate the effect of each bus organization on communication efficiency. How do different bus structures influence system performance and reduce communication bottlenecks?

4. A programmer is developing an industrial automation application and must choose the most suitable instruction format and addressing mode for faster execution and efficient memory access. Construct a concept map linking Instruction Formats, Immediate, Direct, Register, Indirect, and Indexed Addressing Modes with Data Transfer, Arithmetic, and Control Instructions. Show how each addressing mode affects execution speed and memory access. How do instruction formats and addressing modes influence program execution and processor efficiency?

5. An embedded systems engineer is comparing the 8085 and 8086 microprocessors for a factory monitoring system that requires efficient instruction execution and memory organization. Construct a concept map linking their instruction sets, register organization, addressing modes, and memory organization, including memory segmentation in 8086. Show how these features affect program execution and system performance. How do the architectural differences between the 8085 and 8086 influence execution efficiency and memory management?

Code of Conduct

I K·Maheshbabu(192525122) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K·Maheshbabu

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Q. No	Concept Identification (25)	Linkages (25)	Organization (20)	Integration of Literature (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/25	/ 20	/ 20	/ 10	/ 100

Course Coordinator **Faculty Incharge**

MODERN PROCESSOR ARCHITECTURES

KEY ARCHITECTURES

ARM CORTEX
- RISC Architecture
- low power consumption
- used in Mobile devices
- High Energy Efficiency

INTEL CORE
- High performance
- complex Instruction set (CISC)
- Hyper Thereading Technology
- used in desktops, laptops, servers

AMD RYZEN
- Multi-core processing
- High clock speeds
- Good price to performance Ratio
- used in PCs, Workstations

OTHER ARCHITECTURE
- Apple M-series
- RISC-V
- Designed for specialised applications
- Improving performance & Efficiency.

INSTRUCTION EXECUTION PIPELINE

FETCH → DECODE → EXECUTE → MEMORY → WRITE BACK

KEY COMPONENTS

CPU
- Control unit
- ALU
- Registers

Cache memory
- L1 cache
- L2 cache
- L3 cache
- Reduces Memory access time

main memory
- stores Data and instructions
- Higher capacity
- slower than cache

I/o system
- I/o devices
- Interfacing using i/o controllers

Bus structure
- Address bus
- Data Bus
- Control Bus

FACTORS AFFECTING PERFORMANCE

Clock speed
- Higher clock speed leads to better Performance

CPI
- Lower CPI means better performance

Parallelism
- Multi-core and Multi threading improve performance

Cache Efficiency
- Higher cache Hit Ratio improves Performance

APPLICATION AREAS

- mobile devices
- desktop & laptops
- Gaming systems
- Servers & cloud
- H PC

2. Binary and Hexadecimal Number systems

NUMBER SYSTEMS

Binary Number system

- Base = 2
- digits = 0, 1
- uses 2^n weights
- easy for Computers

USES

- Memory organisation
- Instruction Representation
- CPU internals
- Data communication

Hexadecimal Number system

- Base=16
- digits = 0-9 A-F
- uses 16^n weights
- Compact Representation

uses

- Memory Addresses
- Machine Code Representation
- Debugging
- Color codes

Relationship

1 Hexadecimal digit = 4 binary digits

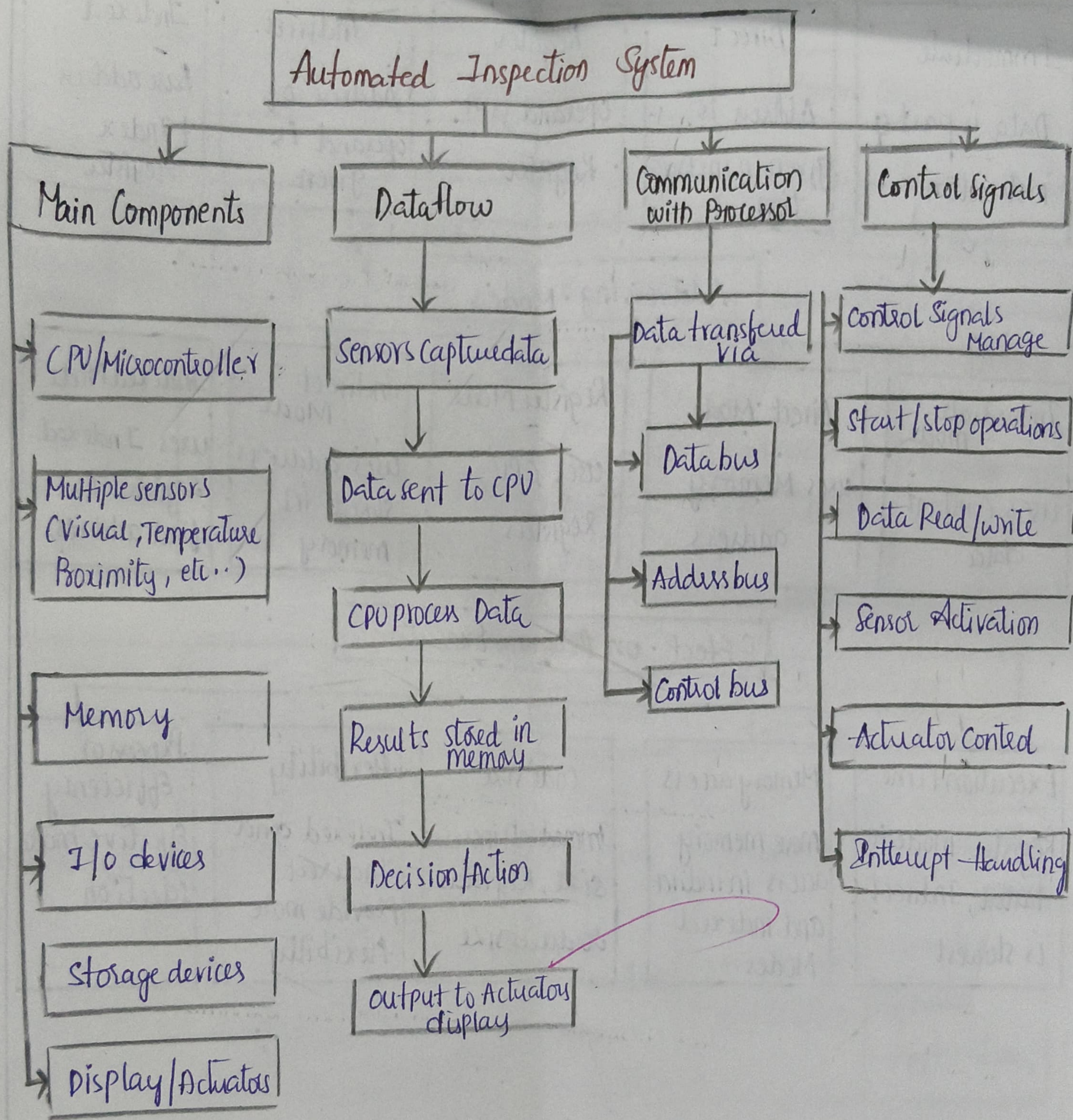
CONVERSIONS

- Binary → Hexadecimal (Group 4 bits and converts)
- Hexadecimal to binary: Replace each hex digit with 4 bits
- Binary → decimal: use 2^n weights and add
- Hexadecimal to decimal: use 16^n weights and add

BENEFIT

Simplify memory organisation, instruction representation and debugging

Automated Inspection system with Multiple Sensors



Benefits

- High inspection Accuracy
- Reduced Human Error
- Real-Time Decision Making
- Faster processing
- Efficient Communication
- Improved Productivity

4.

Instruction format

- **Immediate** — Data is part of instruction
- **Direct** — Address is given in instruction
- **Register** — Operand is in Register
- **Indirect** — Address of operand is given
- **Indexed** — Base address + Index register

Addressing Modes

- **Immediate mode** — uses constant data
- **Direct Mode** — uses Memory address
- **Register Mode** — uses CPU registers
- **Indirect Mode** — uses address stored in memory
- **Indexed Mode** — uses Indexed register to find address

Effect on Program Execution

- **Execution Time** — Register Mode is fastest, Indirect is slowest
- **Memory access** — More memory access in indirect and indexed Modes
- **Code size** — Immediate increases size, Registers reduces size
- **Flexibility** — Indexed and Indirect Provide more flexibility
- **Processor Efficiency** — Effective mode selection

SUMMARY

Proper instruction format and addressing mode selection reduces Execution time, Memory usage and improves overall Processor efficiency.

Aim:

8085 and 8086 Microprocessors - Architecture Comparision.

COMPARISON

Architecture

8085	8086
• 8 bit Micro processor	• 16 bit Micro processor
• single bus system	• seperate Bus
• 16-bit Address Bus	• 20 bit Address bus

Registers

8085	8086
• 8 bit Registers (A, B, C, D, E, H, L)	• 16 bit General purpose Registers (AX, BX, CX, DX) (SI, DI, BP, SP)
• 16 bit Register pair (BC, DE, HL, SP)	• Segment Registers

Memory Organisation

8085	8086
• can address 64 KB memory	• Can address 1MB Memory
• Memory mapped I/O	• seperate I/O space (64KB)

Instruction set

8085	8086
• 74 instructions	• 111 instruction
• simple instruction set	• complex Instruction set
• 1byte/2byte instruction	• 1 to 6 byte instruction

Performance

8085	8086
• Less powerful	• More powerful
• slower Execution	• Faster Execution
• Lower Throughput	• Higher Throughput

Applications

8085	8086
• Embedded systems	• PCs, workstation
• small control applications	• complex software Applications

Conclusion

8086 is more advanced with larger address space, more registers Seperate buses and higher performance compared to 8085